

# PROJECT DOCUMENTATION

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## Entertainment Domain – Netflix Content Analysis

### 1. Executive Summary

This project demonstrates an end-to-end Data Analytics workflow using a Netflix content dataset. The objective is to profile, clean, transform, model, analyze, and visualize the data using Microsoft Excel, Power Query, and Power BI. The final deliverable is an interactive business intelligence dashboard supported by documented insights and recommendations.

### 2. Business Problem

Netflix maintains a large catalog of movies and TV shows from multiple countries, genres, and release years. Data quality issues such as missing values, duplicate records, inconsistent formatting, and invalid entries reduce the reliability of reporting and decision-making. This project addresses these issues through structured data preparation and analytical reporting.

### 3. Project Objectives

- Assess and profile the raw dataset.
- Clean and standardize the dataset using Power Query.
- Build a structured analytical data model in Power BI.
- Develop DAX measures and KPIs.
- Analyze trends across genres, countries, release years, ratings, and content types.
- Design an interactive dashboard to support business decisions.

### 4. Business Questions

1. How is the content library distributed between Movies and TV Shows?
2. Which genres and countries contribute the most content?
3. How has the content library evolved over time?
4. What are the most common content ratings?
5. Which genres have the highest average IMDb ratings?
6. Which directors contribute the largest number of titles?
7. Which regions generate the highest watch hours?
8. What insights can support future content strategy?

## 5. Stakeholders

- Executive Leadership
- Content Strategy Team
- Content Acquisition Team
- Marketing Team
- Business Intelligence & Data Analytics Team

## 6. Project Scope

### In Scope

- CSV data import and profiling
- Data quality assessment
- Data cleaning and transformation
- Data modeling
- DAX measure development
- Dashboard creation
- Business insights and recommendations

### Out of Scope

- Machine Learning
- Predictive Analytics
- Streaming recommendation algorithms
- Real-time data processing

## 7. Deliverables

| Deliverable         | Description                                |
|---------------------|--------------------------------------------|
| Clean Dataset       | Validated and standardized Netflix dataset |
| Data Quality Report | Summary of issues identified               |
| Transformation Log  | Power Query transformation steps           |
| Power BI Model      | Structured relationships and model         |
| Dashboard           | Interactive visual analytics               |
| Insights Report     | Business findings                          |
| Recommendations     | Data-driven recommendations                |
| Documentation       | Complete project documentation             |

## 8. Technology Stack

- Microsoft Excel
- Power Query
- Microsoft Power BI
- DAX

## 9. Success Criteria

- All critical data quality issues resolved.
- Reliable analytical data model created.
- Business questions answered through visualizations.
- Professional dashboard delivered.
- Actionable insights and recommendations documented.

## 10. Project Timeline

| Phase | Activity                        | Status  |
|-------|---------------------------------|---------|
| 1     | Project Documentation           | Planned |
| 2     | Data Collection & Understanding | Planned |
| 3     | Excel Data Profiling            | Planned |
| 4     | Power Query Data Cleaning       | Planned |
| 5     | Data Modeling                   | Planned |
| 6     | DAX Development                 | Planned |
| 7     | Dashboard Development           | Planned |
| 8     | Insights & Recommendations      | Planned |
| 9     | Final Documentation             | Planned |